

```
'Chapter 11. Where to go and what to study next, and how to build a  
chatbot.']
```

We'll continue by *fitting* our TFIDF vectorizer to these chapter descriptions, which will do the document processing to get us ready to create TFIDF vectors whenever we wish. We don't have to do this manually, since there's a `fit()` method defined in the `scikit-learn` module:

```
vctrz.fit(alldocuments)
```

Now, we'll create TFIDF vectors for our chapter descriptions and for a new query asking for a chapter about sorting and searching:

```
query = 'I want to read about how to search for items.'  
tfidf_reports = vctrz.transform(alldocuments).todense()  
tfidf_question = vctrz.transform([query]).todense()
```

Our new query is a natural English language text about searching. The next two lines use the built-in `transform()` and `todense()` methods to create the TFIDF vectors for the chapter descriptions and the query.

Now we have converted our chapter descriptions and query into numeric TFIDF vectors. Our simple chatbot will work by comparing the query TFIDF vector to the chapter description TFIDF vectors, concluding that the chapter the user is looking for is the one whose description vector most closely matches the query vector.

Vector Similarity

We'll decide whether any two vectors are similar with a method called *cosine similarity*. If you've studied a lot of geometry, you'll know that for any two numeric vectors, we can calculate the angle between them. The rules of geometry enable us to calculate angles between vectors not only in two and three dimensions, but also in four, five, or any number of dimensions. If the vectors are very similar to each other, the angle between them will be quite small. If the vectors are very different, the angle will be large. It's strange to think that we can compare English language texts by finding the "angle" between them, but this is precisely why we created our numeric TFIDF vectors—so that we can use numeric tools like angle comparison for data that doesn't start out numeric.